American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

PROJECT TITLE

A Software Engineering Project Submitted
By

Semester: **Fall_24_25**		**Section:**	**Group Number:**	
SN	Student Name	Student ID	Contribution (CO3+CO4+CO5)	Individual Marks
1	Abrar Rejwan Prince	22-48812-3	20%	
2	Faiyaz Azmaine	22-49320-3	20%	
3	Samiul Rayan Rifat	21-45816-3	20%	
4	MD. Ashfak Uzzaman Khan Limon	22-47801-2	20%	
5	Most. Sumaia Farzana Nitu	22-48243-2	20%	

The project will be evaluated for the following Course Outcomes

CO3: ***Select*** **appropriate software engineering models, project management roles, and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects**	Total Marks	
Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]	
Evidence of Argumentation Regarding Process Model Selection	[5Marks]	
Analysis of the impact of societal, health, safety, legal, and cultural issues	[5Marks]	
Submission, Defense, Completeness, Spelling, grammar, and Organization of the Project report	[5Marks]	
CO4: ***Develop*** **a project management plan to manage software engineering projects following the principles of engineering management and economic decision process**	Total Marks	
Develop the project plan, its components of the proposed software products	[5Marks]	
Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]	
Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	

CO5: Perform as an effective team member or leader in diverse team settings and solve multi-disciplinary problems in the computer science and engineering domain	Total Marks	
Taking project responsibility: perform assigned tasks on time independently	[5 Marks]	
Contribution to project group meetings, sharing fruitful ideas	[5Marks]	
Positive attitude towards group work, collaboration, compromise, helping others to understand their project work responsibility	[5Marks]	
Showing respect and value towards other team member's opinion	[5Marks]	

Description of Student's Contribution in the Project work

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Contribution in Percentage (%):
Contribution in the Project:

- Contribution Description 1
- Contribution Description 2

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Signature of the Student

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Contribution in the Project:

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- Contribution Description 2

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Signature of the Student

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Contribution in the Project:

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Signature of the Student

1. PROJECT PROPOSAL

1.1 Background to the Problem

In recent decades, emergency services like fire service, providing ambulance, law enforcement service etc. have become a necessity for everyday human life. Emergency services have evolved quite a lot in the past years by pushing the boundaries and incorporating technological innovations. Emergency services are captivating due to their potential to save lives. The challenge lies in minimizing the response time, providing these services in the least amount of time from the time of requesting the service and managing these services simultaneously. The problem arises from the great amount of time emergency services are provided from the time of the request and people facing consequences because of that. Ease of use, automation and solid software like interface is essential to overcome these challenges and provide emergency services in the least amount of time. The problem is super important because it affects our ability to be safe and to have faith in cases of emergencies. Emergency services can teach us how important someone's time can be in such cases. When we create an efficient and innovative system and deploy it to help people to have fire, police and ambulance services at the palm of their hands we learn more about software, computer, human computer interaction and how they can help us everywhere, not just in emergency segments. In simpler words, solving this problem helps us to save lives more efficiently than ever before.

1.2 Solution to the Problem

The traditional emergency service system in Bangladesh faces significant limitations that hinder its efficiency and responsiveness. The current infrastructure can only handle 100 calls simultaneously and process a maximum of 30,000 calls per day, leading to long queues and delayed service. This constraint increases the waiting time for emergency seekers, defeating the primary purpose of providing immediate assistance in critical situations. Additionally, the system relies heavily on manual handling, which slows down communication between requesters and service providers, increasing the risk of life-threatening delays.

1.3 Proposed Solution

The proposed software aims to revolutionize the emergency response system by integrating advanced technologies to overcome existing limitations. It will streamline request management, minimize response times, and enhance the user experience, ensuring faster and more reliable services. By leveraging artificial intelligence, data analytics, and real-time communication tools, this system will provide a seamless and efficient bridge between emergency service seekers and providers.

Key Features and Benefits

1. **AI-Powered Request Management**: Automates request handling, prioritizing emergency cases using historical data and predictive analysis.

2. **Real-Time Communication and GPS Integration**: Enables instant tracking of requesters' locations to expedite response times.

3. **User-Friendly Interface**: Simplifies request submission without phone calls, making the service more accessible and efficient.

4. **Compact and Cost-Effective Design**: A lightweight, optimized system that reduces hardware costs and operational expenses.

5. **Enhanced Autonomy and Efficiency**: Minimizes human intervention, streamlining the process to deliver faster, more reliable emergency services.

6. **User Friendly Fast Response UI** : Allow user to interact with the AI and determine the problem as fast as possible.

By implementing these features, the system will eliminate bottlenecks, reduce dependency on manual operations, and enhance overall performance. This innovative solution aims to save lives by providing faster, more secure, and scalable emergency services that inspire trust and confidence in public safety infrastructure.

1.4 Basic Functionalities

User Authentication: Secure login with phone numbers, passwords, fingerprints, and OTP verification; signup with personal details and location permissions.

Request Management: Request fire brigade, healthcare, or law enforcement services, with options to specify urgency, cause, and severity.

Real-Time Tracking: Track emergency response units in real-time and access past location data for enhanced transparency.

AI-Based Request Analysis: Generates probable causes of requests using historical data for faster and more accurate responses.

After-Service Feedback: Allows users to review services, improving quality and accountability.

Misuse Prevention: Generates penalties for false requests to discourage abuse and ensure service availability.

1.5 Target Users

The proposed emergency response management system is designed to serve:

- General Public and Emergency Seekers: Individuals requiring immediate assistance for emergencies such as medical issues, fires, or crimes.
- Emergency Responders and Service Providers: Police, medical teams, and fire departments responsible for delivering emergency services.
- Government Agencies and Dispatch Personnel: Authorities and operators managing emergency service coordination and infrastructure.

This comprehensive approach ensures a streamlined, user-centric system that improves service efficiency and public safety.

2. SOFTWARE DEVELOPMENT LIFE CYCLE

2.1 Process Model

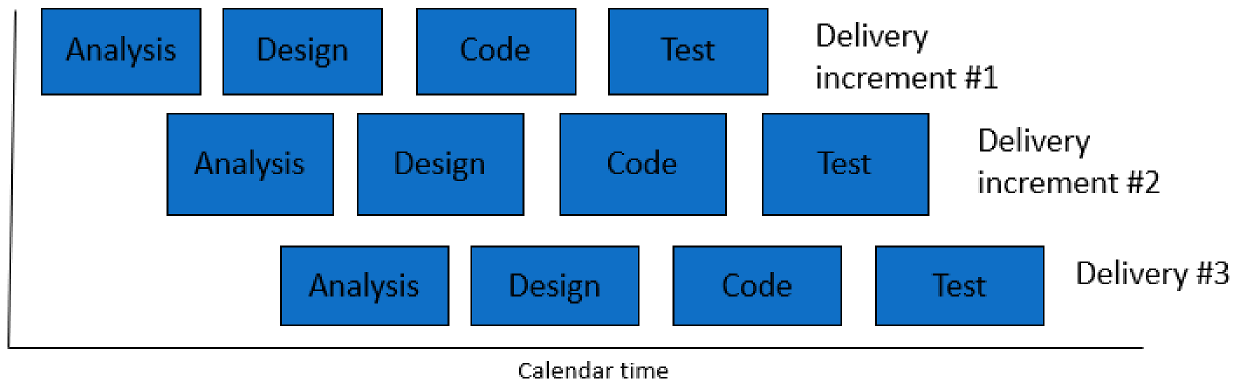
The Incremental Development Model is the best approach for our emergency service system project because it offers several key benefits that match our project's needs.

This model helps manage the complexity of our system. Since our project involves multiple features like real-time location tracking, request handling, secure login, and penalty management, the Incremental Model lets us develop and test each feature separately. For example, we can work on location tracking as one part, service requests as another, and penalty management as a third. This allows us to release each feature step-by-step and improve them as we go, making the development process more manageable.

The Incremental Model also allows us to get feedback throughout the project. Since our system will be used in high-pressure emergency situations, it's important that we make updates based on real-world feedback from users and responders. Each feature can be improved as we progress, ensuring it meets actual needs and works well in real scenarios.

It is crucial to deliver core features early to reduce risks. The Incremental Model helps us launch essential functions, such as login, request handling, and tracking, in the first stages. This means the system will be useful to users right away, even while we develop more advanced features later on. By delivering in stages, we avoid the risks of waiting for the entire system to be finished before going live.

The Incremental Model helps with team and resource management. We can divide the project into smaller parts, so different teams can work on different features at the same time. This makes the development process faster and more efficient, allowing each team to focus on what they do best. This parallel approach makes the Incremental Model a practical and effective choice for our complex project.



Incremental Development

- Provide an analysis regarding the nature and environment of the software that you are going to develop and select the best suitable method(s) to develop the software.
- Present your arguments based on your analysis about why your selected method(s) is the best choice among all other methods to develop your proposed software.
- Presents sufficient amount of evidence to support argument for your model selection in developing your proposed solution.

2.2 Project Role Identification and Responsibilities

Project Manager: The Project Manager provides coordination and oversight for iterative cycles of planning, development, and review. They ensure that resources, schedules, and scope are iteratively adjusted to accommodate evolving stakeholder input and requirements. The Project Manager primarily focuses on overseeing the incremental delivery of functional components while keeping in step with the project's big-picture objectives.

Business Analyst: The Business Analyst continuously elaborates on the project's requirements through close collaboration with stakeholders. The priorities of features in each iteration are determined based on client feedback and changing requirements. The Business Analyst ensures that the team stays aligned with the project's goals and that any changed requirements are properly documented throughout the process.

System Architect/Designer: The System Architect/Designer creates a system architecture to support iterative development. They design modular, scalable components that can be incrementally improved. At the end of each cycle, the designs and interfaces are updated to reflect findings and feedback, ensuring the system remains responsive to emerging requirements.

Development Team: The Development Team is responsible for delivering functional components through small iterative cycles. They develop features, integrate feedback from clients and testers into the code, and ensure it remains maintainable and adaptable for future increments. Their work focuses on achieving high-quality outcomes after each incremental delivery.

Test Team and Quality Assurance: After every iteration, the Test Team performs rigorous testing to ensure the functionality works as expected and identifies any bugs. They provide valuable feedback to the Development Team for defect resolution and ensure each incremental delivery meets the specified requirements. Their iterative testing ensures consistent quality and functionality improvement of the product.

Documentation Team: This team develops and updates project-related documentation throughout the system's lifecycle, including technical documents, user manuals, and specifications. Following each iteration, they document changes and enhancements to ensure the evolving system is well-documented and accessible.

Customer/Client: The Client is actively engaged in reviewing iterative deliverables to support continuous improvement. They approve incremental deliveries, provide insights for refinement, and collaborate with the project team to reassess priorities as the project unfolds.

Safety and Compliance Experts: These professionals continuously review the project to ensure every aspect of the process complies with safety and legal standards. As new features or components are developed, they update safety protocols and compliance measures to ensure adherence to the latest standards throughout the iterative process. All roles emphasize key aspects such as flexibility, continuous feedback, and incremental improvement. This approach allows better adaptation to changing needs while ensuring collaboration and regulatory compliance.

Existing emergency service solutions in Bangladesh rely on manual, call-based systems with limited capacity, lacking real-time tracking, predictive analytics, and integration across services, leading to delays in response. Our proposed solution introduces an AI-powered unified platform with real-time GPS tracking, predictive analytics, and a user-friendly interface, ensuring faster responses, better resource management, and improved accessibility, addressing current gaps effectively.

2.3 Impact on Social

The development of an emergency service system as described addresses significant societal, health, safety, legal, and cultural issues, enhancing public well-being and trust. Here is a detailed analysis of these impacts:

Societal Impact: The proposed system bridges critical gaps in emergency response infrastructure. By enabling real-time tracking, quick access to services, and AI-based decision-making, it improves community resilience to crises. Efficient emergency handling fosters public confidence, minimizes response disparities in urban and rural areas, and reduces societal disruptions during emergencies.

Health Impact: Faster access to healthcare services, including ambulances and virtual consultations, can significantly lower mortality and morbidity rates. The system's streamlined response mechanism ensures immediate medical attention, thereby addressing life-threatening conditions more effectively and reducing the burden on healthcare systems.
Safety Enhancements: Incorporating precise location tracking and reducing response times greatly enhances personal and public safety. Automated alert systems prevent human error, and penalties for false reports deter misuse, ensuring that resources remain available for genuine emergencies.
Legal Considerations: The system must comply with data privacy and cybersecurity regulations, given access to personal data and real-time location information. Secure login protocols, user verification, and proper data management ensure adherence to local laws while protecting user rights. False request penalties also align with legal frameworks designed to prevent service misuse.
Cultural Sensitivity: Emergency services are critical across diverse cultural contexts. Customizable service preferences, language support, and region-specific emergency categorization promote inclusivity and accessibility, ensuring equitable service delivery. Tailoring services to cultural nuances helps build trust and ensures widespread adoption.

Overall, this integrated approach strengthens emergency management, aligns with legal standards, enhances public health and safety, and respects cultural diversity, making it a comprehensive solution to modern emergency response challenges.

3. Functional Requirements

1. Software Login

1.1 The software shall allow users to log in with their phone number and password.

1.2 The login credentials (phone number and password) will be verified within database records of registered users.

1.3 If correct credentials have been provided, an one-time password (OTP) will be sent to the users registered phone number to verify the ownership of the account.

1.4 If successfully logged in, the interactive home page of the user account will be displayed.

1.5 If wrong login credentials (phone number and password/ fingerprint) have been inserted, then a randomized captcha will appear to make sure the attempt was not made by an AI.

1.6 If 5 times wrong login credentials (phone number and password/ fingerprint) have been inserted, any further login attempt from the machine in question will be locked in an hour. The time will multiply by 3 on every set of wrong login attempts.

Priority Level: High

Precondition: Users have valid and registered phone numbers and password.

2. Software Signup

2.1 The software allows users to register with their phone number, password along with other information (name, dob, nid no. etc.).

2.2 The software shall ask the users to accept their terms and conditions.

2.3 The software shall verify the ownership of the phone number through a one-time password (OTP) sent to the registered phone number.

2.4 If successfully verified, the interactive home page of the user will be displayed.

2.5 The software shall prompt the users to give permission of the location service or the global positioning system (gps) of the user's machine.

2.6 If location permission is given, the software shall let the users choose whether they want to add their insurance papers now or later, so that the charged amount for the emergency services gets covered by insurance agencies.

2.7 The software shall let the users add any available payment method (Bank transfer, mobile banking, online banking etc). The users can choose to add payment method later, then the default payment method will be added to the users account which is cash on delivery (COD).

2.8 If location permission is rejected, the software shall prompt the users to that location access is necessary to receive emergency services.

Priority Level: High Precondition: User have valid phone number.

3. Generate Probable Causes of Requests

3.1 The software shall fetch the information from the databases of the emergency services (fire brigade, health department, law enforcement department).

3.2 The software shall record all of the causes of the requests.

3.3 The software shall use all the fetched information and the recorded data generate multiple probably causes of the requests based on the geographical location of the users by using data analytics and artificial intelligence (AI).

Priority Level: High

Precondition: User has been logged in successfully into the software.

4. Request Fire Brigade

4.1 The software shall let the users request a fire brigade unit.

4.2 The software shall generate probable causes of the request.

4.3 The software shall let the users choose the cause of their request from the generated probable causes.

4.4 The software shall let the users call the emergency number if the generated probable causes doesn't match with the causes of the users' requests to let the emergency responders know the details about the emergency.

4.5 If the users select any one from the probable causes then the software shall let the users choose how severe the emergency is from three predefined options (high, medium, low).

4.6 The software shall then ask the users for the final confirmation of the request showing the estimated time of arrival.

4.7 If confirmed, the fire brigade unit will be notified immediately with the users' information and based on the severity of the emergency one or multiple units will be assigned.

4.8 The software will keep on showing the real time location of the fire brigade and the location of the incident inside a well-coordinated map until the fire brigade units have successfully reached the location of the incident.

4.9 If cancelled, the software shall return the users to the home page.

Priority Level: High

Precondition: User has been successfully logged into the software.

5. Request Health Care Service

5.1 The software shall allow the users to request for health care services.

5.2 The software shall then ask the users to choose between two options (nearby ambulance, emergency consultant). If nearby ambulance has been selected by the users, The software shall allow the users to choose from the category of ambulances (premium, moderate, basic).

5.3 The software shall then ask the users to enter the name of the hospital they want to go.

5.4 The software shall then select the closest ambulance of the selected category within a 1km radius from the location of the users.

5.5 The software shall assign the request to the ambulance driver.

5.6 The software shall then show the real time location of both the ambulance and the requester along with the estimated time of arrival and the charged amount.

5.7 If an emergency consultant is selected in , the software shall allow the users to choose from the available specialists.

5.8 The software shall then connect the requester with the selected specialist through a video call.

Priority Level: High

Precondition: User has been successfully logged into the software.

6. Request Law Enforcement Unit

6.1 The software shall let the users to request for law enforcement unit(s).

6.2 The software shall generate probable causes of the request.

6.3 The software shall let the users choose the cause of their request from the generated probable causes.

6.4 The software shall let the users call the emergency number if the generated probable causes doesn't match with the causes of the users' requests to let the emergency responders know the details about the emergency.

6.5 If the users select any one from the probable causes then the software shall let the users choose how severe the emergency is from three predefined options (high, medium, low).

6.6 The software shall then ask the users for the final confirmation of the request showing the estimated time of arrival.

6.7 If confirmed, the nearest law enforcement unit will be notified immediately with the users' information and based on the severity of the emergency one or multiple units will be assigned.

6.8 The software will keep on showing the real time location of the law enforcement unit and the users' location inside a well-coordinated map until the law enforcement unit(s) have successfully reached the users' location.

6.9 If cancelled, the software shall return the users to the home page.

Priority Level: High

Precondition: User has been successfully logged into the software.

7. After Service Review

7.1 After the service have been provided successfully, the software shall ask the users to give a review of the responders and vice versa.

7.2 This review will be added to the accounts of the users.

7.3 False requests will be penalized and an amount will be charged on the next request of the users in question.

7.4 The penalty will increase 5 times on each false request.

7.5 If no false request of fraudulent activity were found within 3 years then the amount will be set to the initial amount.

Priority Level: High

Precondition: User has a verified account registered on the software.

8. Track Last and Realtime Location

8.1 The software shall acquire the permission to access the global positioning system embedded into the user's system.

8.2 The software shall continuously monitor real-time location during active emergency services.

8.3 The software shall provide live map interface for the users and emergency responders.

8.4 The software shall handle any inaccuracies due to factors like signal interference.

8.5 The software shall retrieve and display last known location to the emergency responders if any connection problem occurs.

Priority Level: High

Precondition: User has a verified account registered on the software and has given location access to the software.

9. Generate Penalties

9.1 The software shall ask the users to give their honest feedback or review of the service provider after the successful completion of the service.

9.2 The software shall then take the review and send it to the emergency service provider's client and store the information into the database for future evaluation.

9.3 The software shall then ask the service providing units to give their honest feedback and comment on the legitimacy of the users.

9.4 These reviews will be added together with the users accounts and points will be generated based on the review (Negative points for negative reviews and positive points for positive reviews).

9.5 Based on these points, if enough negative point has been added to the users accounts, the software shall incorporate an pre stated amount for penalizing the users.

9.6 The software shall stop the user from requesting the services before the penalized amount has been paid.

Priority Level: High

Precondition: User has a verified account registered on the software. User has successfully taken any listed service.

10. Track Last and Realtime Location

1. The software shall acquire permission to access the global positioning system embedded into the user's system.

2. The software shall continuously monitor real-time location during active emergency services.

3. The software shall provide live map interface for the users and emergency responders.

4. The software shall handle any inaccuracies due to factors like signal interference.

5. the software can retrieve and display the last known location to the emergency responders if any connection problem occurs.

Priority Level: High

Precondition: User has a verified account registered on the software and has given location access to the software.

4. Non-functional Requirement

1. Response Time

1.1 The system shall respond to user inputs (such as login, request for services, or location updates) within 2-3 seconds under normal network conditions.
1.2 The software shall ensure a loading time below 1 second for frequently accessed pages, such as the home page and real-time map.
1.3 All backend processing (e.g., data analytics, AI-generated probable causes) shall complete within 5 seconds.
1.4 If a delay occurs due to network issues, the system shall show a progress indicator and offer the user options to retry.

Priority Level: High
Precondition: User has an active internet connection.

2. Availability

2.1 The system shall maintain an uptime of 99.99% over a given month.
2.2 The software shall support auto-recovery mechanisms in the event of a crash or unexpected shutdown.
2.3 The system shall include fallback mechanisms (like cached data) to ensure continued operation even during temporary backend service outages.
2.4 Emergency services (fire, health, law enforcement) must remain operational at all times, including during maintenance windows.

Priority Level: High
Precondition: The backend server infrastructure is properly configured and maintained.

3. Authentication

3.1 The software shall enforce secure login using phone number and password, followed by a One-Time Password (OTP) verification.
3.2 The software shall support Two-Factor Authentication (2FA) via SMS or authenticator app for added security.
3.3 All credentials shall be stored using industry-standard encryption and hashing methods (e.g., SHA-256, crypt).
3.4 Session tokens shall expire after 15 minutes of inactivity to ensure security.
3.5 Any suspicious login activity shall trigger a verification alert to the user's registered contact method.

Priority Level: High
Precondition: User has a registered account with verified credentials and access to the registered phone number.

4. Scalability

4.1 The system shall support a minimum of 10,000 concurrent users during peak traffic hours.
4.2 The backend architecture shall be designed using microservices or modular services for horizontal scalability.
4.3 The system shall automatically allocate resources (e.g., cloud instances, containers) to handle increased user loads.
4.4 The database shall use sharding or replication to ensure smooth performance during high query loads.
4.5 Load balancers shall be employed to distribute traffic evenly across server nodes.

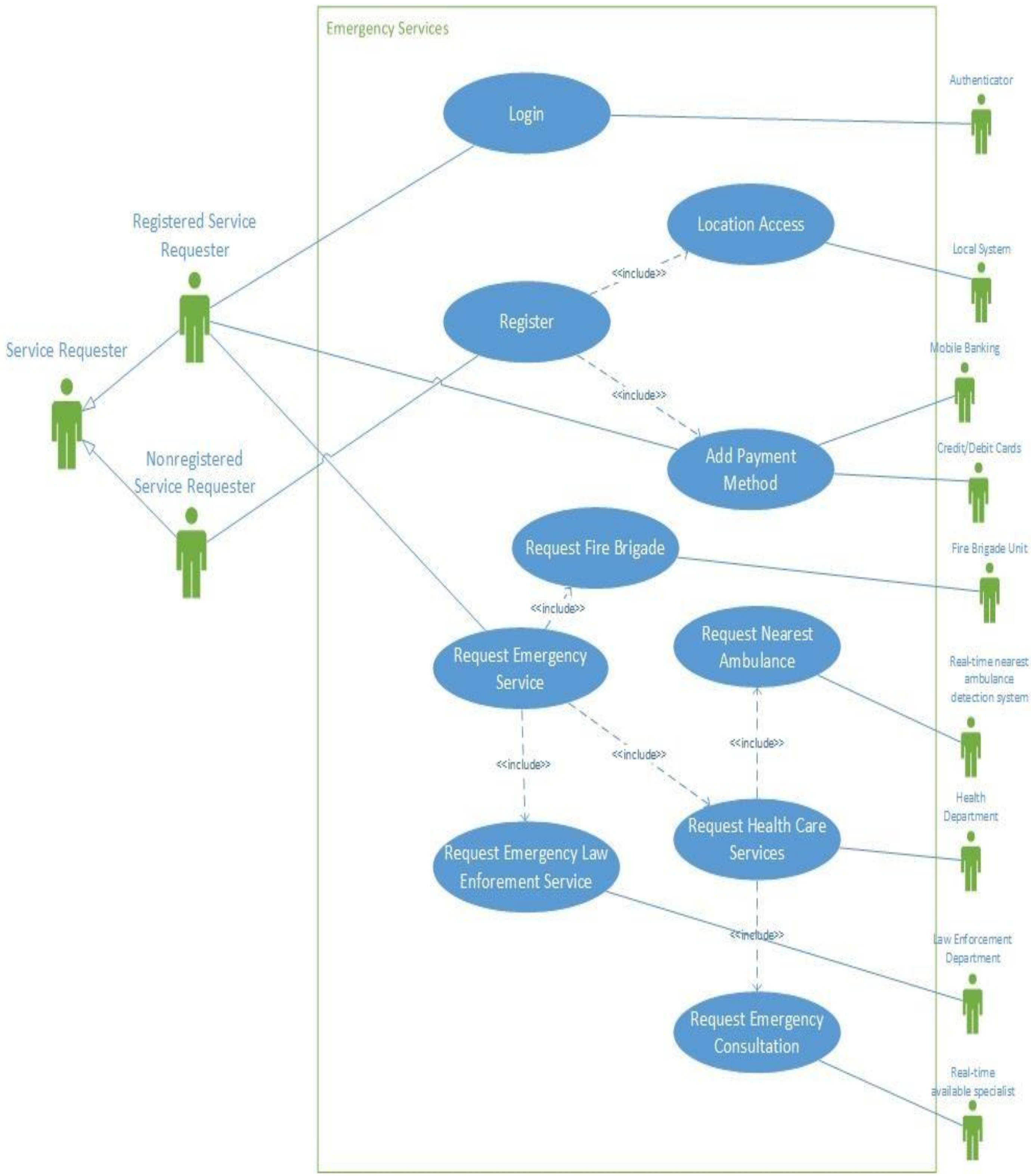
Priority Level: High
Precondition: System is deployed on a cloud infrastructure (e.g., AWS, Azure, GCP) with autoscaling enabled.

5. Platform Independence

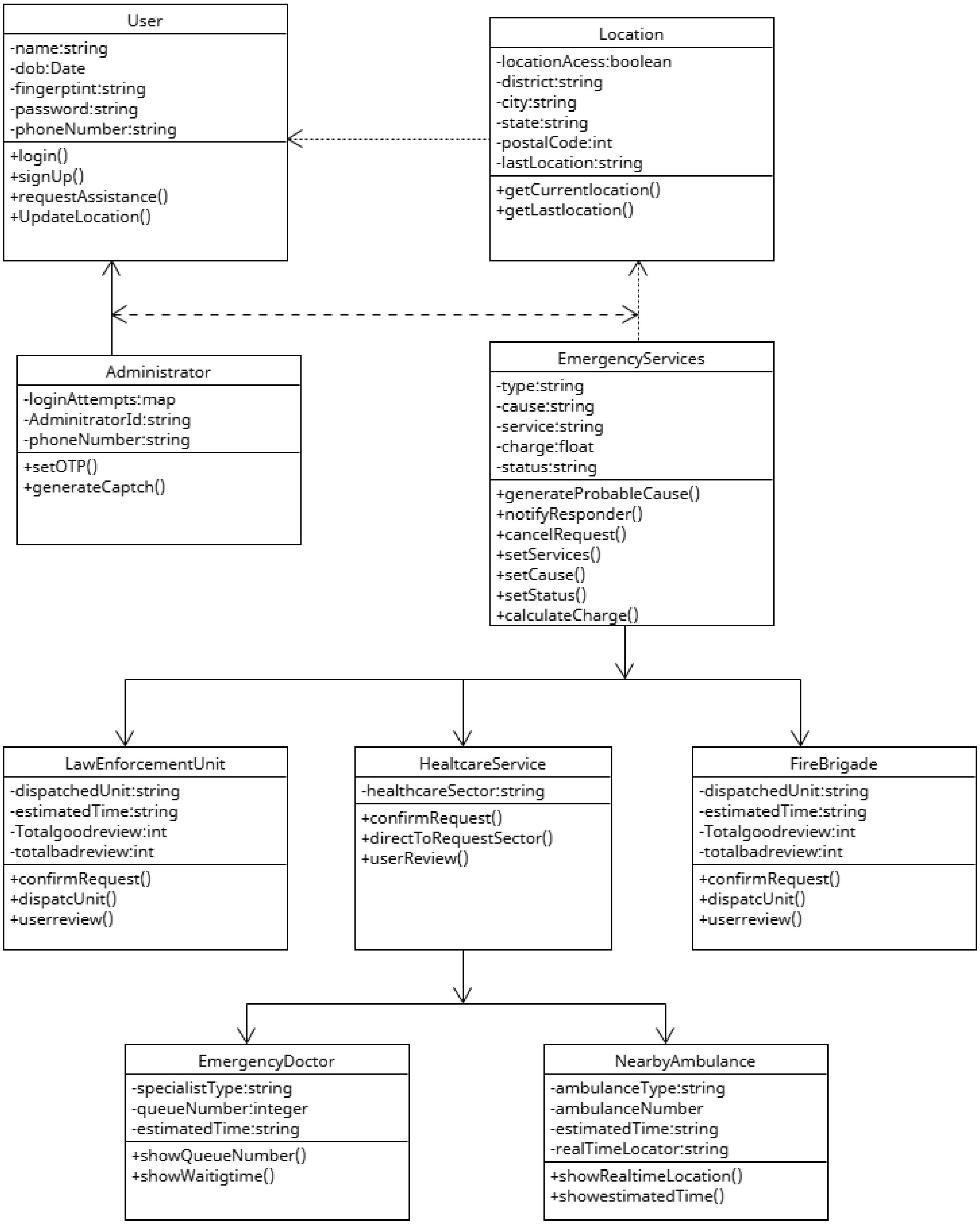
5.1 The software shall be fully compatible with all major operating systems (Android, iOS, Windows, macOS).
5.2 A responsive web application shall be provided alongside mobile apps, ensuring accessibility via any modern browser.
5.3 The codebase shall be developed using cross-platform technologies (e.g., Flutter, React Native).
5.4 The UI/UX experience shall remain consistent across all platforms with optimized layouts and controls.
5.5 All core features (location tracking, requests, authentication) must be functional regardless of the user's platform.

Priority level: High
Precondition: The device must meet the minimum software and hardware requirements and have an active internet connection.

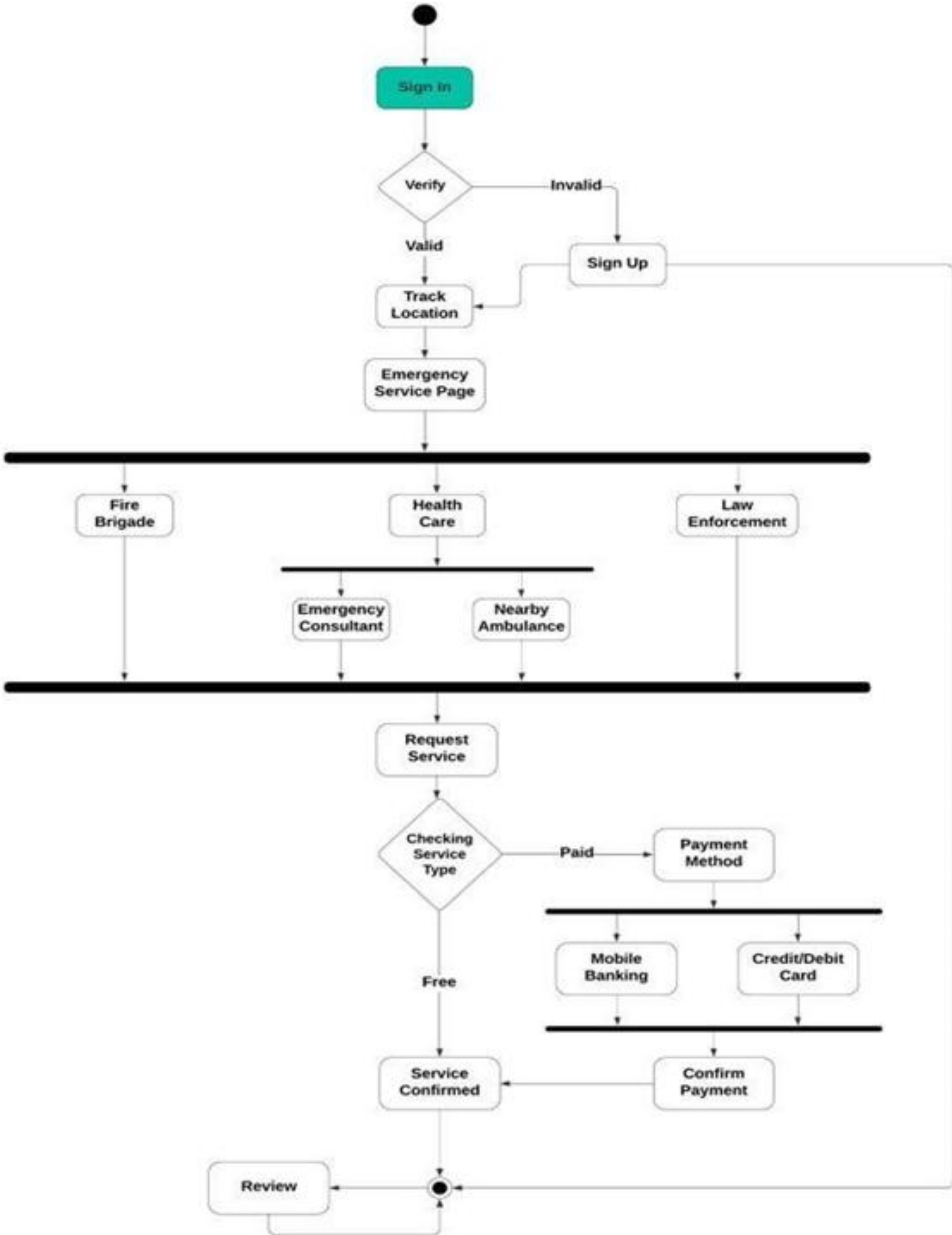
Use case Diagram



Class Diagram



Activity Diagram



Sequence Diagram